A Homogeneous Compensation Branch for Indirect Evolutionary Rescue in a Reduced Free-Space Predator–Prey Model

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Abstract

Indirect evolutionary rescue can occur when adaptive change in one species prevents extinction of another. We analyze this problem in a deterministic reduced free-space predator–prey model with evolving prey defense and imposed predator mortality stress. The supported mechanism is a homogeneous compensation branch: higher stress is offset by lower equilibrium prey defense $q^*(s)$, which increases predator conversion opportunity and preserves a positive predator equilibrium. A frozen-q counterfactual initialized at the pre-stress branch state verifies a deterministic rescue window, approximately $0.0726 \leq s \leq 0.1656$, in which fixed defense loses predators while evolving defense preserves them. A stricter threshold-safe ecological window, approximately $0.0726 \leq s \leq 0.1511$, excludes deterministic recoveries after very low predator density. Sampled spatial diagnostics and finite-amplitude perturbation follow-up did not provide evidence for persistent spatial-pattern-mediated rescue in the tested reduced setup. Controlled nonlinear trade-off extensions recover the linear branch and support compensation for selected concave, convex, and mixed shapes, but remain local and parameter-sensitive. The final interpretation is model-specific: indirect evolutionary rescue here is driven by homogeneous compensation over the verified counterfactual window.

Keywords: indirect evolutionary rescue; eco-evolutionary dynamics; predator–prey model; reaction–diffusion PDE; trade-off; Routh–Hurwitz stability

1 Introduction

Evolutionary rescue describes persistence after environmental deterioration through adaptive change [Gomulkiewicz and Holt, 1995, Bell and Gonzalez, 2009, Bell, 2017]. In predator–prey systems, rescue can also be indirect: evolution in prey may alter predator persistence. In the setting studied here, prey defense reduces predation but carries ecological costs. When predator mortality stress increases, predator pressure declines, selection can shift prey toward lower defense, and the lower-defense prey state can increase predator conversion opportunity. This feedback is the qualitative mechanism of indirect evolutionary rescue introduced by Yamamichi and Miner [2015], not a concept introduced here. Community-level and predator-mortality follow-up work has examined related rescue or response mechanisms in changing multispecies systems [Fussmann and Gonzalez, 2013, Cortez and Yamamichi, 2019, Hermann and Becks, 2022].

Prey defense evolution is a natural target for this question because the defense trait changes both prey ecological performance and predator energetic gain. Rapid eco-evolutionary feedbacks in predator–prey systems can alter ecological dynamics on short timescales [Yoshida et al., 2003]. Recent experimental and modeling work also shows that prey trait variation can change the relative evolutionary contribution to predator growth rates [Hermann et al., 2024]. The exact form of the defense trade-off is important for that reason: nonlinear or asymmetric trade-offs can change predator–prey eco-evolutionary dynamics [Kasada et al., 2014]. In this manuscript, trade-off geometry is therefore part of the mechanism rather than a background detail.

Spatial structure is a plausible competing explanation because local dispersal and heterogeneous predator–prey overlap can change persistence outcomes. Spatial eco-evolutionary feedbacks can mediate coexistence in prey–predator systems [Colombo et al., 2019]. A scalar persistence threshold is an attractive first diagnostic: one compares the mortality stress at which predators persist or fail in the ODE and PDE. That threshold framing can fail, however, when long transients, continuation dependence, or multiple reachable outcomes occur at the same stress. In that case, the correct object is not a single threshold but the mechanism organizing the reachable basins.

The mechanism supported by the present analysis is not spatial-pattern-mediated rescue. Fixed-defense spatial tests in this model sequence did not provide evidence for robust predator rescue. The well-mixed eco-evolutionary ODE supports indirect evolutionary rescue over the frozen-q counterfactual window identified below. Subsequent ODE–PDE comparison showed that the basin structure observed in the PDE is primarily inherited from homogeneous reaction dynamics rather than generated by persistent spatial patterning. The important mechanism is therefore a homogeneous compensation branch, with spatial analyses used to test whether the PDE destabilizes or qualitatively alters that branch.

Indirect evolutionary rescue itself is therefore already established, especially by Yamamichi and Miner. The contribution here is narrower and model-specific. We isolate an explicit homogeneous compensation branch in a reduced free-space predator–prey model, verify its role with a frozen-defense counterfactual, and then ask whether the tested spatial PDE replaces this reaction-level mechanism by persistent spatial-pattern-mediated rescue. The contribution is mechanism separation within a reduced deterministic model.

This manuscript analyzes the mechanism organizing predator persistence in the tested model. We derive the compensation branch, state its feasibility conditions, evaluate local stability using Routh–Hurwitz criteria, test spatial-mode stability in the PDE, evaluate targeted finite-amplitude heterogeneous perturbations, and examine controlled nonlinear trade-off extensions. The claim hierarchy is intentionally conservative. Strong claims are restricted to the tested ODE branch, its feasibility and local stability, the tested PDE spatial-mode stability calculation, targeted non-homogeneous perturbation outcomes, and the controlled nonlinear shape grid. Broader claims about global stability, all trade-off forms, all diffusion coefficients, or all heterogeneous initial conditions are not made.

We first formulate the ODE and PDE models. We then derive the homogeneous compensation branch and its feasibility conditions. We analyze local stability using Routh-Hurwitz criteria, test spatial-mode stability in the PDE, evaluate targeted non-homogeneous perturbations, and finally examine controlled nonlinear trade-off extensions.

2 Model

Relation to Roy/Sasmal defended-prey models

Roy/Sasmal defended-prey free-space models study interactions among free space, defended prey, undefended prey, predators, and spatial pattern formation in reaction–diffusion systems [Sasmal et al., 2019, Roy et al., 2026]. The present manuscript builds on that model family but asks a different mechanism question: whether spatial structure mediates indirect evolutionary rescue when predator mortality stress is imposed externally. The formulation here is therefore a reduced (n, w, q, z) free-space formulation, not an algebraically equivalent reduction of the full Roy PDE. For spatially homogeneous states, one may heuristically identify $u = (1-q)n$ and $v = qn$. This identification is not used as an exact PDE reduction because separate diffusion of u and v would generally generate additional composition-gradient terms. Table 1 summarizes the conceptual mapping.

Table 1: Relationship between Roy/Sasmal defended-prey variables and the reduced formulation used here. The mapping is conceptual: q summarizes prey composition, and q-diffusion is a phenomenological closure rather than an algebraic derivation from separate defended and undefended prey PDEs.

Full Roy/Sasmal variables	Reduced variables	Biological meaning	Retained or collapsed role
u, v	n, q	Undefended and defended prey; total prey density and defended-prey frequency	Total prey density is retained as n; prey composition is summarized by q; separate u and v density fields are collapsed
w	w	Predator density	Predator density is retained, with externally imposed mortality stress s used to test rescue
z	$z = \kappa^{-1} - n - w$	Free space	Free-space limitation is retained through the algebraic constraint used in prey growth and interaction terms
Spatial diffusion of u, v, w	Diffusion of n, w, q	Spatial movement or smoothing	n and w diffusion are retained phenomenologically; q-diffusion represents defense-frequency composition smoothing, not a derived full-u, v closure

The model is a reduced free-space eco-evolutionary formulation inspired by defended and undefended prey–predator reaction–diffusion models with free space and aposematic defense trade-offs [Sasmal et al., 2019, Roy et al., 2026]. Let $n(t)$ denote prey density, $w(t)$ predator density, $q(t) \in [0, 1]$ defended-prey frequency, and

$$z = \kappa^{-1} - n - w$$

the free-space variable. The variable z represents unused ecological capacity or free space that modulates local prey recruitment and predator conversion opportunity. The biologically admissible state space is $n \geq 0$, $w \geq 0$, $z \geq 0$, and $0 \leq q \leq 1$. The variable s denotes externally imposed predator-mortality stress. For the linear trade-off model,

$$r(q) = r_u(1-q) + r_v q, \quad a(q) = a_u(1-q) + a_v q, \quad b(q) = b_u(1-q) + b_v q.$$

Here $r(q)$ is the prey growth trade-off, $a(q)$ is the attack or palatability trade-off, and $b(q)$ is the predator energetic-gain or conversion-opportunity trade-off. The term $b(q)$ affects predator growth directly; it is not a direct term in prey selection except through the ecological feedbacks represented by the full system. Subscripts u and v denote undefended and defended endpoint values, respectively. With these variables, the well-mixed ODE is

$$\frac{dn}{dt} = n[r(q)z - \xi - a(q)w], \tag{1}$$
$$\frac{dw}{dt} = w[b(q)nz - (m+s) - \mu w], \tag{2}$$
$$\frac{dq}{dt} = \nu q(1-q)[(r_v - r_u)z - (a_v - a_u)w]. \tag{3}$$

In the prey equation, $r(q)z$ is free-space-limited prey recruitment, ξ is background prey loss, and $a(q)w$ is predation loss. In the predator equation, $b(q)nz$ is predator energetic gain or conversion opportunity, $(m + s)$ is baseline plus imposed predator mortality, and μw is density-dependent predator loss. In the q-equation, the bracket is the marginal effect of increasing defended-prey frequency on prey per-capita growth, and the $q(1 - q)$ factor is the standard boundary-preserving frequency factor that keeps $q = 0$ and $q = 1$ invariant in the ODE. Parameter values are listed in Table 2.

The spatial PDE uses the same local reaction terms and diffusion coefficients D_n, D_w, and D_q for prey density, predator density, and defense-frequency smoothing:

$$\partial_t n = D_n \nabla^2 n + n[r(q)z - \xi - a(q)w], \tag{4}$$

$$\partial_t w = D_w \nabla^2 w + w[b(q)nz - (m + s) - \mu w], \tag{5}$$

$$\partial_t q = D_q \nabla^2 q + \nu q(1 - q)[(r_v - r_u)z - (a_v - a_u)w]. \tag{6}$$

The PDE tests use a rectangular spatial domain $[0, L_x] \times [0, L_y]$ with zero-flux Neumann boundary conditions. The q-diffusion term is a phenomenological composition-smoothing closure for the defense-frequency variable. It is not derived here from separate defended and undefended prey density equations, and that modeling choice could affect PDE spatial-stability conclusions. Because q is a frequency variable, the numerical implementation enforces admissibility by clipping q to $[0, 1]$ after explicit PDE updates. This clipping is a numerical admissibility constraint, not a proof that the phenomenological PDE closure preserves $q \in [0, 1]$ without enforcement. Finite-amplitude PDE conclusions should therefore be interpreted under this implementation. The homogeneous ODE compensation branch and frozen-q counterfactual do not depend on D_q, because all diffusion terms vanish on spatially constant states.

For the controlled nonlinear trade-off extension, endpoint values are preserved but linear interpolation is replaced by shape functions:

$$r(q) = r_u + (r_v - r_u)f_r(q), \quad a(q) = a_u + (a_v - a_u)f_a(q), \quad b(q) = b_u + (b_v - b_u)f_b(q),$$

with $f_\gamma(q) = q^\gamma$, an endpoint-preserving shape function used to vary trade-off curvature. This nonlinear form is used only for controlled extension tests, not to replace the baseline linear model.

3 Methods and numerical protocols

The analysis combines targeted numerical integration with analytic or semi-analytic calculations. All numerical values reported in this manuscript are taken from CSV outputs in the repository. The reported analyses consist of ODE no-evolution counterfactual integrations, dense linear-branch stability checks, saved-field audits, linear algebraic modal scans, and targeted finite-amplitude PDE perturbation tests; they do not introduce broad PDE scans or model-equation changes.

ODE analysis. ODE trajectories were integrated from specified initial means for representative trajectories and q_0–w_0 basin maps. ODE integrations used `scipy.integrate.solve_ivp` with the LSODA method unless otherwise noted. For the no-evolution counterfactual, the horizon was $T = 3000$, with 1201 saved time points, relative tolerance 10^{-8}, absolute tolerance 10^{-10}, extinction threshold 10^{-4}, and a terminal tail window equal to the final 25 percent of the trajectory. The threshold is a deterministic diagnostic/quasi-extinction convention used to distinguish threshold-safe trajectories from deterministic recoveries after very low predator density; it is not an empirical extinction probability. Persistent, extinct, and transient labels were assigned from late-time predator density, terminal behavior of $q(t)$, relative change between late windows, and residual-style steady-state diagnostics. Predator-persistent outcomes have positive predator density

Table 2: Parameter values used in the compensation-branch and spatial-stability analyses. These values define the tested mechanistic case study; they are not empirical estimates.

Symbol	Meaning	Value used	Role in model
r_u	Growth contribution of undefended prey	1	Baseline prey growth endpoint
r_v	Growth contribution of defended prey	0.65	Defense-growth trade-off endpoint
a_u	Attack or palatability endpoint for undefended prey	1	Baseline predation endpoint
a_v	Attack or palatability endpoint for defended prey	0.35	Defense-palatability trade-off endpoint
b_u	Conversion endpoint for undefended prey	0.08	Predator conversion opportunity at low defense
b_v	Conversion endpoint for defended prey	0.02	Predator conversion opportunity at high defense
κ	Inverse habitat capacity scale	0.15	Sets free-space relation $z = \kappa^{-1} - n - w$
ξ	Prey loss parameter	0.55	Appears in prey balance
m	Baseline predator mortality	0.1	Predator mortality before added stress
μ	Predator density-dependent loss	0.2	Predator self-limitation term
ν	Defense evolution rate	0.05	Scales q-dynamics
D_n	Prey diffusion coefficient	0.01	PDE prey movement
D_w	Predator diffusion coefficient	0.01	PDE predator movement
D_q	Defense-frequency diffusion coefficient	0.005	PDE defense-frequency smoothing
L_x	Domain length in x	20	Neumann spatial-mode domain scale
L_y	Domain length in y	20	Neumann spatial-mode domain scale

with sufficiently small terminal change. Extinct outcomes have predator density near zero. Transient outcomes do not settle clearly into either class within the tested horizon. Exact classification thresholds and source CSV mappings are given in Supplementary Section S1.

We distinguish two persistence notions. *Deterministic tail-based persistence* is the baseline mathematical classifier: a trajectory can be persistent if its late-time tail has positive predator density and settles, even if it dipped below the extinction threshold earlier. *Threshold-safe persistence* additionally requires that $w(t)$ never cross below the extinction threshold during the tested horizon. The latter is more conservative for ecological interpretation because a biological population that falls below an extinction threshold may not recover even if the deterministic ODE later returns to a positive equilibrium.

Compensation branch computation. For the linear trade-off model, the compensation branch was derived analytically from the interior equilibrium equations, giving closed-form z^*, w^*, n^*, and $q^*(s)$. Numerical equilibria from multiple initial guesses were used as consistency checks against the analytic branch. For nonlinear trade-offs, the branch was parameterized by q through $c(q) = r'(q)/a'(q)$ and the corresponding stress map $s(q)$.

No-evolution counterfactual. To verify that predator persistence depends on prey evolution rather than only on the existence of a positive branch, we compared the full evolving-q ODE with a frozen-q counterfactual. Both treatments were initialized at the pre-stress branch state $(n^*, w^*, q^*(0))$, with $q^*(0) \approx 0.6726$. In the counterfactual treatment $dq/dt = 0$, so q remained fixed at this pre-stress value. The frozen-q run asks whether the same ecological stress would eliminate predators if prey composition could not evolve away from its pre-stress value. Stress was swept across the interior branch interval, and predator persistence/extinction was classified using the same tail-window logic as the ODE analyses. The verified rescue window is the finite-grid, finite-horizon interval where frozen-q trajectories are extinct while evolving-q trajectories persist. As a complementary analytic check, for fixed $q_0 = q^*(0)$, the prey-only frozen-q equilibrium satisfies

$$z_0 = \frac{\xi}{r(q_0)}, \qquad n_0 = \kappa^{-1} - z_0,$$

and the predator invasion boundary is

$$s_{\text{fixed}} = b(q_0) n_0 z_0 - m.$$

For the current parameterization, $s_{\text{fixed}} \approx 0.0696$, above which predators cannot invade the prey-only state when defense is fixed. In the tested finite-horizon trajectories initialized from the pre-stress branch state, predator loss occurs just above this threshold.

Local stability. Local stability of the linear branch was evaluated with the analytic reaction Jacobian. For the cubic characteristic polynomial $\lambda^3 + A_1\lambda^2 + A_2\lambda + A_3 = 0$, the Routh–Hurwitz inequalities $A_1 > 0$, $A_2 > 0$, $A_3 > 0$, and $A_1A_2 > A_3$ determined local stability. Finite-difference Jacobians were used for selected nonlinear trade-off extensions where closed-form expressions were not the main object.

PDE spatial stability. Homogeneous ODE equilibria are homogeneous PDE steady states because diffusion vanishes on spatially constant fields. Spatial-mode stability was evaluated from $J_F(U^*) - \lambda_{ij} D$, where $J_F(U^*)$ is the reaction Jacobian, $D = \mathrm{diag}(D_n, D_w, D_q)$, and λ_{ij} are Neumann eigenvalues on the rectangular domain. The zero mode recovers ODE stability; nonzero modes test infinitesimal spatial perturbations. Discrete Neumann modes were evaluated for $i, j = 0, \ldots, 64$, and a continuous λ-scan sampled 600 equally spaced values over the corresponding mode-equivalent λ-range. A denser modal Routh–Hurwitz diagnostic sampled 2001 λ values over the same interval and computed the cubic coefficients of $J_F(U^*) - \lambda D$. Because this is a sampled modal calculation rather than a symbolic proof for all λ, the spatial-stability claim is stated as dense sampled evidence over the tested range.

Non-homogeneous PDE perturbations. Targeted finite-amplitude PDE tests used localized predator patches, localized defense patches, sinusoidal modes, random heterogeneity, and basin-boundary heterogeneity. PDE tests used 64×64 grids on 20×20 domains with $D_n = 0.01$, $D_w = 0.01$, $D_q = 0.005$, and explicit timestep $dt = 0.1$, with CFL checks in the simulation helper. Perturbation amplitudes were targeted rather than exhaustive: local predator patches used $A = 0.5$, local defense patches used $A_q = 0.2$, sinusoidal and random heterogeneity used $A = A_q = 0.1$, and basin-boundary heterogeneity used $A = A_q = 0.25$. Random seeds included 20260702 and 20260703 where stochastic heterogeneity was used. Each heterogeneous run was compared with a matched homogeneous control at the same stress and baseline mean state. Basin-changing heterogeneous cases were followed to longer horizons to distinguish persistent basin changes from finite-horizon transient labels. A supplemental audit reports the min/max q values in saved diagnostic fields, so the bounds statement refers to saved post-update fields under the clipped implementation rather than an unclipped scheme.

Nonlinear trade-off extension. The nonlinear extension used the endpoint-preserving shape family $f_\gamma(q) = q^\gamma$ with $\gamma \in \{0.5, 1, 2\}$ for the growth, attack, and conversion trade-offs. This is a controlled local shape grid, not a broad scan over all nonlinear trade-off functions.

Table 3: Numerical summary for the baseline compensation-branch analysis. Values are taken from the script-generated CSV outputs.

Quantity	Value	Interpretation
z^*	1.1917	Free-space value along the linear branch
n^*	4.8333	Prey density along the linear branch
w^*	0.6417	Predator density along the linear branch
$q^*(0)$	0.6726	Pre-stress branch defense frequency
s_{fixed}	0.0696	Frozen-q prey-only predator-invasion threshold
Interior branch interval	$(-0.1131, 0.2324)$	Feasibility interval for $0 < q^*(s) < 1$
Deterministic rescue window	[0.0726, 0.1656]	Frozen-q extinct, evolving-q tail-persistent
Threshold-safe rescue window	[0.0726, 0.1511]	Evolving-q does not cross $w = 10^{-4}$
Extinction threshold	10^{-4}	Predator-density threshold used in classifiers
Baseline ν	0.05	Defense-evolution rate in the manuscript parameterization
PDE domain	20×20	Neumann spatial domain used for finite-amplitude PDE tests

4 Homogeneous compensation branch

Define

$$\Delta r = r_v - r_u, \quad \Delta a = a_v - a_u, \quad \Delta b = b_v - b_u.$$

These endpoint differences measure how defended and undefended endpoint values differ in growth, palatability, and conversion. For an interior equilibrium with $n > 0$, $w > 0$, and $0 < q < 1$, the nontrivial equilibrium conditions are

$$r(q)z - \xi - a(q)w = 0,$$

$$b(q)nz - (m + s) - \mu w = 0,$$

$$\Delta r z - \Delta a w = 0.$$

At an interior trait equilibrium with $0 < q < 1$, the prefactor $\nu q(1 - q)$ is nonzero, so $dq/dt = 0$ requires the selection-gradient bracket to vanish. If

$$c = \frac{\Delta r}{\Delta a},$$

then the selection-gradient condition gives

$$w^* = cz^*.$$

The ratio c is therefore fixed by trade-off geometry through the selection-gradient equation; it measures how the growth cost of defense trades against the predation-reduction benefit in the prey selection gradient and is not an additional fitted parameter. Because the trade-offs are linear, $r(q) - ca(q) = r_u - ca_u$, so the prey-balance equation gives

$$z^* = \frac{\xi}{r_u - ca_u}, \quad w^* = cz^*, \quad n^* = \kappa^{-1} - z^* - w^*.$$

The predator-balance equation then determines the defense frequency required at stress s:

$$q^*(s) = \frac{(m + s + \mu w^*)/(n^* z^*) - b_u}{b_v - b_u}.$$

Because n^*, w^*, and z^* are fixed by the prey-balance and selection-gradient equilibrium conditions in the linear case, increasing stress s is absorbed by changing $q^*(s)$ through the predator-balance equation. This branch explains rescue as compensation: increasing predator mortality stress is offset by decreasing equilibrium prey defense frequency, which increases predator conversion opportunity and keeps a positive predator equilibrium feasible. In the current parameterization, the branch has approximately $n^* \approx 4.8333$ and $w^* \approx 0.6417$, while $q^*(s)$ decreases with stress.

No-evolution counterfactual. The counterfactual test starts at the pre-stress branch state and compares the full evolving-q dynamics with a frozen-q treatment in which $q = q^*(0)$ is held fixed. The frozen-q prey-only invasion calculation gives $s_{\text{fixed}} \approx 0.0696$, above which predators cannot invade the prey-only state when defense is fixed. In the tested finite-horizon trajectories initialized from the pre-stress branch state, predator loss occurs just above this threshold. On the finite targeted stress grid and integration horizon, the deterministic tail-based rescue window is approximately

$$s \in [0.0726, 0.1656],$$

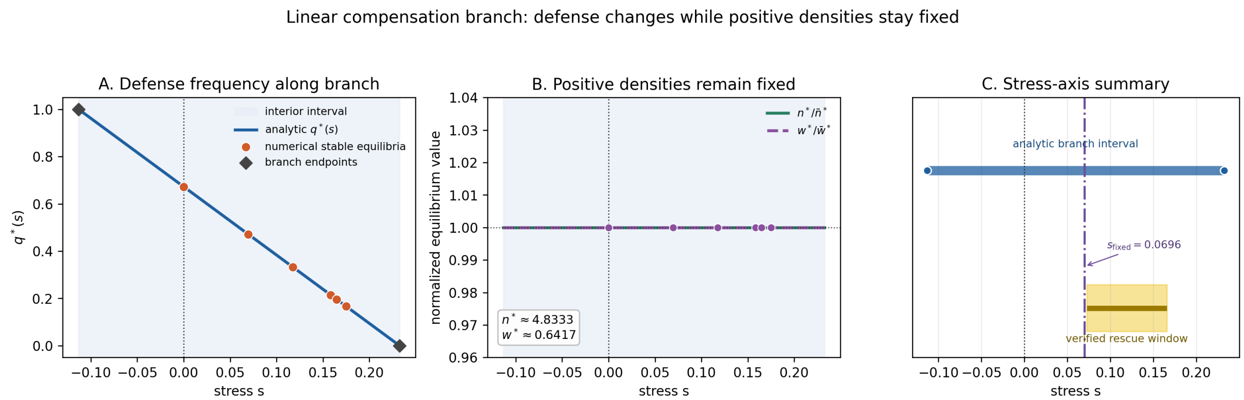

Figure 1: Compensation branch and stress-axis summary. Panel A plots analytic $q^*(s)$ against numerical stable persistent equilibria. Panel B shows normalized n^* and w^*, which remain effectively fixed along the linear branch. Panel C places the analytic branch interval, the frozen-q invasion threshold s_{fixed}, and the verified finite-grid rescue window on a common stress axis.

where frozen-q trajectories are classified as predator-extinct while evolving-q trajectories remain predator-persistent by the baseline tail-window classifier. Requiring threshold-safe persistence gives a narrower window,

$$s \in [0.0726, 0.1511],$$

because some deterministic trajectories recover after dipping below $w = 10^{-4}$. At the representative stress $s = 0.1191$, the evolving-q trajectory lowers defense and maintains $w \approx 0.6417$, whereas the frozen-q trajectory loses the predator. This supports the indirect evolutionary rescue interpretation over the verified counterfactual window, with the threshold-safe window as the more conservative ecological version. The window is a reachable-window result from the pre-stress branch initial condition, not the full analytic feasibility interval of the interior branch. Outside this window, especially near the upper branch boundary, the pre-stress initial condition can fall outside the persistent basin or cross the extinction threshold even though the analytic branch remains feasible. The supplemental q_0–w_0 basin map documents this basin dependence.

Boundary and predator-free equilibria. For fixed $q = q_0$ and $w = 0$, the prey-only equilibrium satisfies

$$z_0 = \frac{\xi}{r(q_0)}, \qquad n_0 = \kappa^{-1} - z_0,$$

provided $n_0 > 0$. A rare predator grows when

$$b(q_0)n_0z_0 - (m + s) > 0,$$

so the frozen-q invasion boundary is $s_{\text{fixed}} = b(q_0)n_0z_0 - m$. With $q_0 = q^*(0)$, this gives $s_{\text{fixed}} \approx 0.0696$. The analytic interior branch endpoints are different objects: they occur when $q^*(s)$ reaches $q = 1$ or $q = 0$, which are trait-boundary feasibility endpoints of the interior formula. We do not classify those endpoints as saddle-node or transcritical bifurcations. The upper edge of the observed rescue window is therefore interpreted as a basin/reachability and finite-time classification issue from the pre-stress initial condition, not simply as loss of branch feasibility.

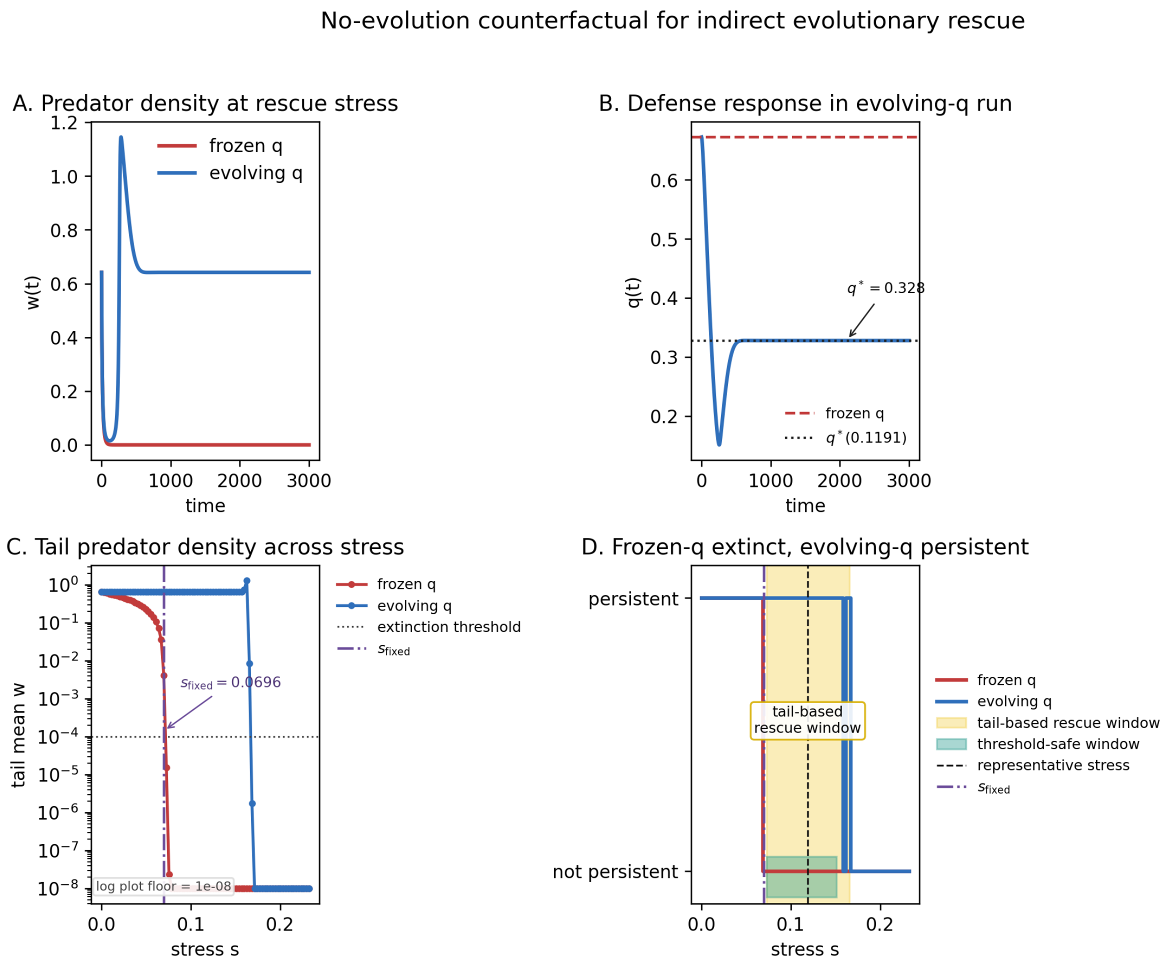

Figure 2: No-evolution counterfactual for the linear ODE. Panel A compares predator density at a representative rescue stress for evolving-q and frozen-q treatments. Panel B shows the defense response in the evolving-q run. Panel C plots tail mean predator density across stress for both treatments; values below 10^{-8} are shown at the plotting floor for log-scale visualization. Panel D distinguishes the deterministic tail-based rescue window from the narrower threshold-safe window.

5 Branch feasibility and local stability

The interior branch exists only under explicit feasibility conditions:

$$c > 0, \quad r_u - ca_u > 0, \quad z^* > 0, \quad w^* > 0, \quad n^* > 0, \quad 0 < q^*(s) < 1.$$

Solving the branch equation for stress gives

$$s(q) = n^* z^* [b_u + \Delta b q] - m - \mu w^*.$$

Therefore the open stress interval for an interior branch is the interval between the values at $q = 0$ and $q = 1$. In the current parameterization this is approximately

$$s \in (-0.1131, 0.2324).$$

The target stresses used in the mechanism tests lie inside this interval. These conditions make the branch a conditional mathematical statement for the linear trade-off ODE rather than a purely numerical observation.

The endpoints of this analytic interior branch occur when $q^*(s)$ reaches the trait boundaries $q = 0$ or $q = 1$. They are feasibility or trait-boundary endpoints of the interior formula. We do not classify these endpoints as saddle-node, transcritical, or other bifurcations; a full boundary-equilibrium bifurcation analysis is outside the present scope.

Local stability of the homogeneous branch was evaluated with the analytic Jacobian of the ODE reaction system. The coefficients A_1, A_2, and A_3 below are the coefficients of the characteristic polynomial of this reaction Jacobian evaluated at the branch equilibrium. For the three-dimensional system, the characteristic polynomial can be written

$$\lambda^3 + A_1\lambda^2 + A_2\lambda + A_3 = 0.$$

The cubic Routh–Hurwitz conditions for local stability are

$$A_1 > 0, \quad A_2 > 0, \quad A_3 > 0, \quad A_1A_2 > A_3.$$

For the current parameterization, these inequalities hold at all target stresses tested along the branch and agree with direct eigenvalue stability. A dense scan of 1001 stress values across the open interior interval $(-0.1131, 0.2324)$, excluding each trait-boundary endpoint by 10^{-8}, also satisfied the Routh–Hurwitz inequalities and eigenvalue stability. The smallest dense-scan Routh–Hurwitz margin was positive but approached zero near the trait-boundary endpoints. This supports local stability of the sampled open homogeneous compensation branch. It does not prove global attraction, endpoint stability, or all basin boundaries.

6 Spatial PDE stability and finite-amplitude heterogeneity

A homogeneous ODE equilibrium is also a homogeneous PDE steady state because diffusion terms vanish for spatially constant fields. Let $U^* = (n^*, w^*, q^*)$ denote a branch equilibrium, $J_F(U^*)$ the ODE reaction Jacobian evaluated at that state, $D = \mathrm{diag}(D_n, D_w, D_q)$ the diffusion matrix, and λ_{ij} the Neumann eigenvalue for spatial mode (i, j). Linearizing around a homogeneous equilibrium and expanding in Neumann eigenmodes replaces the Laplacian by $-\lambda_{ij}$, giving the modal matrix

$$J_F(U^*) - \lambda_{ij}D,$$

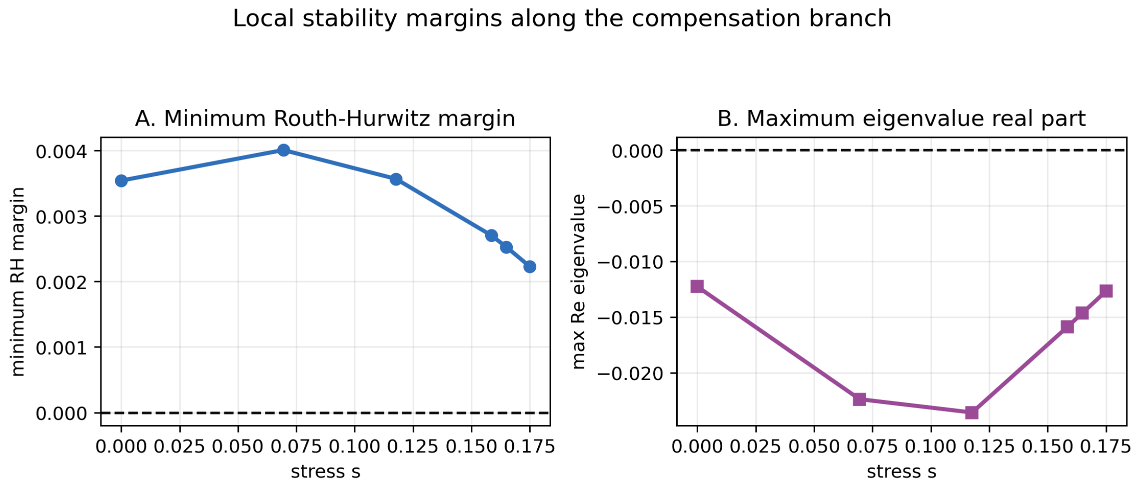

Figure 3: Local stability margins along the compensation branch. Panel A shows the minimum Routh–Hurwitz margin across A_1, A_2, A_3, and $A_1A_2 - A_3$. Panel B shows the maximum real part of the reaction-Jacobian eigenvalues.

where

$$\lambda_{ij} = \left(\frac{i\pi}{L_x}\right)^2 + \left(\frac{j\pi}{L_y}\right)^2.$$

The zero mode is the ODE stability problem. Nonzero modes test whether infinitesimal spatial perturbations grow.

For the current diffusion coefficients, all tested nonzero Neumann modes with $i, j = 0, \ldots, 64$ were stable at the target branch stresses. A sampled continuous-λ scan of $\max \Re\mathrm{eig}(J_F(U^*) - \lambda D)$, using 600 equally spaced λ values over the corresponding mode-equivalent range, also found no positive growth. A dense sampled modal Routh–Hurwitz scan with 2001 λ values over the same range found positive modal margins at all target branch stresses; the smallest sampled modal margin was approximately 2.23×10^{-3}, and the largest sampled modal eigenvalue real part was approximately -1.27×10^{-2}. A controlled diffusion-ratio linear grid found no spatial instability in the tested ratios. Thus the PDE did not destabilize the homogeneous branch through the tested infinitesimal spatial modes and modal range. This is still a tested-range result, not an analytic proof over all spatial modes or all diffusion coefficients.

The q-diffusion closure matters for this spatial interpretation. In the modal matrix, the term $-\lambda D$ directly damps nonzero q-component perturbations through D_q. Therefore the negative spatial-pattern result is conditional on the reduced (n, w, q) closure, the phenomenological q-diffusion term, the clipped q-admissibility implementation, the tested diffusion values, the 20×20 Neumann domain, and the tested finite-amplitude setup. A supplemental linear diagnostic varied $D_q \in \{0, 0.001, 0.005, 0.01, 0.05\}$ with $D_n = D_w = 0.01$; it found no positive sampled modal growth over the tested λ-range, including the $D_q = 0$ case, but this remains a linear diagnostic and not a nonlinear PDE robustness result.

Linear spatial stability is not enough to rule out finite-amplitude heterogeneous effects. Targeted finite-amplitude PDE tests therefore used localized predator patches, localized defense patches, sinusoidal perturbations, smooth random heterogeneity, and basin-boundary heterogeneity. These tests compared heterogeneous fields with matched homogeneous controls. Initial finite-horizon basin-label changes occurred for localized defense perturbations near basin-boundary mean states, but these changes were transient-label effects without persistent spatial patterning. Longer-horizon analysis showed that the basin-changing cases resolved to the matched homogeneous-control basin, with final spatial coefficients of variation below the persistence threshold.

The spatial interpretation is therefore specific: in the tested PDE, spatial heterogeneity can create transient excursions near basin boundaries, but we found no evidence for a persistent spatial-pattern-mediated rescue mechanism.

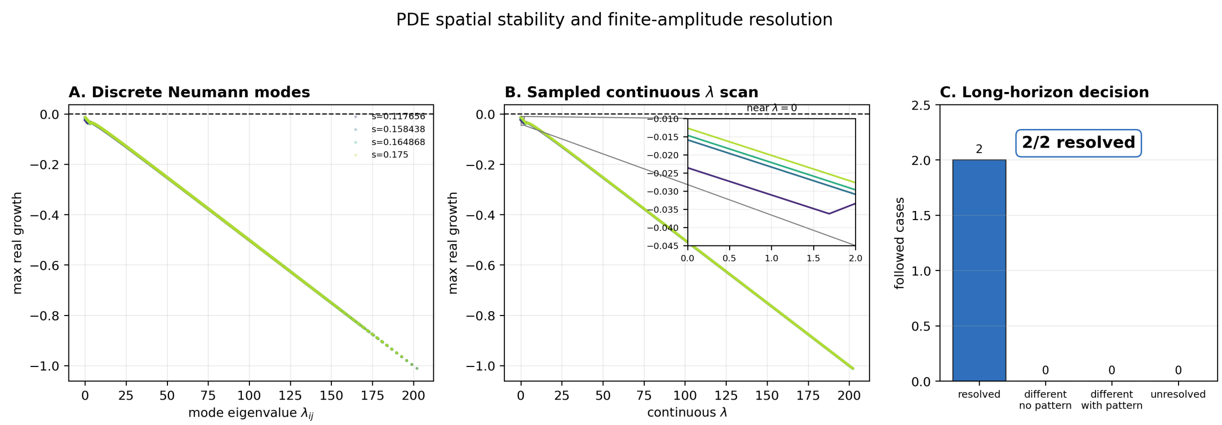

Figure 4: Spatial-mode and finite-amplitude PDE tests. Panel A plots maximum real growth rates for tested discrete Neumann modes, computed from $J_F(U^*) - \lambda_{ij}D$. Panel B shows the sampled continuous-λ scan over 600 equally spaced values in the mode-equivalent range corresponding to $i, j = 0, \ldots, 64$; supplemental modal Routh–Hurwitz margins provide a denser sampled coefficient check over the same interval. Panel C summarizes the two followed finite-amplitude heterogeneous cases that changed finite-horizon basin labels; both resolved to their matched homogeneous controls without persistent spatial patterning.

7 Nonlinear trade-off extension

For nonlinear trade-offs, the branch is most naturally parameterized by q. The extension uses the endpoint-preserving shape function $f_\gamma(q) = q^\gamma$ to vary trade-off curvature while keeping the same endpoint values. When derivatives are defined and $a'(q) \neq 0$, the selection-gradient condition gives

$$c(q) = \frac{r'(q)}{a'(q)}.$$

The corresponding branch geometry is

$$z(q) = \frac{\xi}{r(q) - a(q)c(q)}, \quad w(q) = c(q)z(q), \quad n(q) = \kappa^{-1} - z(q) - w(q),$$

and predator balance maps the branch to stress:

$$s(q) = b(q)n(q)z(q) - m - \mu w(q).$$

The linear branch is recovered when all shape exponents are one.

The controlled nonlinear extension used $\gamma_r, \gamma_a, \gamma_b \in \{0.5, 1, 2\}$, representing concave, linear, and convex endpoint-preserving shapes. This grid is small and structured; it is not a global trade-off scan. This use of controlled nonlinear trade-off shapes follows the broader message that trade-off geometry can qualitatively affect eco-evolutionary predator–prey dynamics [Kasada et al., 2014], but the present grid is only a local mechanism check rather than a broad empirical trade-off survey. Within that grid, the generalized formulation recovered the linear branch, and selected concave, convex, and mixed nonlinear shapes supported feasible locally stable compensation branches. The shape-grid summary recorded 11 robust compensation shapes, 11 partial compensation shapes, 4 no-compensation shapes, and 1 unresolved shape. Targeted PDE spatial-mode and non-homogeneous tests for selected stable nonlinear branches did not provide evidence for persistent spatial-pattern-mediated rescue.

The nonlinear result suggests that compensation is not only a linear-trade-off artifact within this controlled shape grid. It also emphasizes parameter sensitivity: nonlinear shape can change branch feasibility, stability, and stress interval length. Mechanistically, failure can occur if the selection-gradient ratio $c(q) = r'(q)/a'(q)$ fails to generate positive feasible $z(q)$, $w(q)$, and $n(q)$. Compensation can also fail if the stress map $s(q) = b(q)n(q)z(q) - m - \mu w(q)$ does not overlap rescue-relevant stresses. Conversion curvature in $b(q)$ can prevent lower defense from producing enough predator-growth compensation. Finally, branch existence alone is not sufficient: the branch must also be locally stable and reachable from the relevant initial conditions. These statements are mechanistic interpretations of the controlled shape-grid diagnostics, not a general theorem for all nonlinear trade-offs.

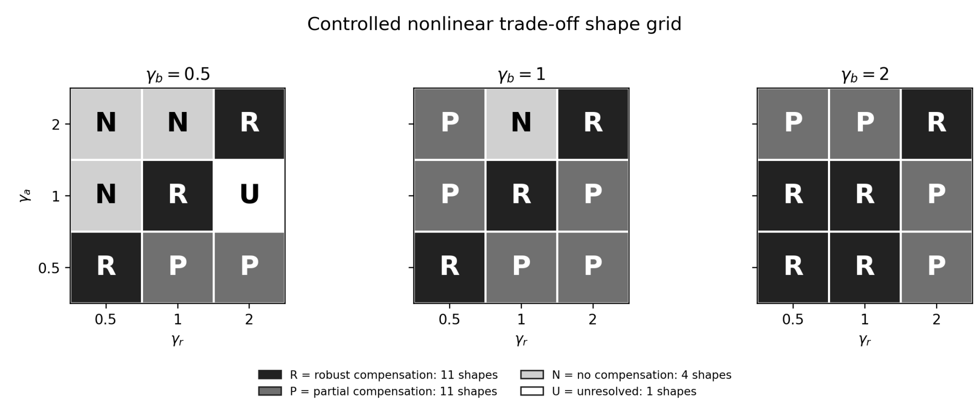

Figure 5: Controlled nonlinear trade-off extension summary. The figure shows the structured shape grid faceted by conversion-shape exponent γ_b, with growth and attack exponents on the axes. Cell categories match the numerical summary: 11 robust compensation shapes, 11 partial compensation shapes, 4 no-compensation shapes, and 1 unresolved shape.

8 Biological interpretation

Higher predator mortality directly reduces predator persistence. In this model, prey defense evolution can compensate because mortality stress reduces predator pressure, and lower predator pressure favors lower prey defense. Lower defense increases predator conversion opportunity through $b(q)$, which can restore predator growth balance at a positive predator equilibrium. The frozen-q counterfactual verifies this as indirect evolutionary rescue on the finite tested stress grid where fixed prey defense leads to predator extinction but evolving prey defense preserves the predator.

The supported biological interpretation is that the homogeneous compensation branch is a reaction-level explanatory mechanism rather than a pattern-mediated one. Defended prey are less favorable to predators in the conversion term, so a shift toward lower defense makes prey more exploitable by the predator. Under the tested trade-off values, that evolutionary shift is sufficient to offset part of the imposed predator mortality stress. The predator persists indirectly because the prey population evolves toward a state that improves predator growth balance in this deterministic parameterized model.

We found no evidence that the tested PDE creates persistent spatial-pattern-mediated rescue. Instead, the PDE preserves the homogeneous compensation branch, which is locally stable to tested spatial modes. Spatial stability matters biologically because it means small spatial variation around the homogeneous rescue state is damped rather than amplified under the current diffusion coefficients. The targeted non-homogeneous tests likewise did not show evidence of persistent spatial patterns, while still allowing transient heterogeneity near basin boundaries.

Figure 6 summarizes this claim hierarchy.

Why pattern formation in Roy/Sasmal models does not imply pattern-mediated rescue here. Roy/Sasmal defended-prey free-space models establish that spatial pattern formation can occur in selected defended-prey predator–prey regimes [Sasmal et al., 2019, Roy et al., 2026]. The present manuscript builds on that literature but asks a different mechanism question: whether spatial structure mediates indirect evolutionary rescue after imposed predator mortality stress in a reduced free-space formulation. In the tested reduced stress model, the evidence supports homogeneous compensation, and targeted spatial diagnostics did not provide evidence for persistent pattern-mediated rescue. This conclusion should not be read as a contradiction of Roy/Sasmal spatial pattern formation or as evidence against Turing or spatio-temporal pattern formation in full defended/undefended prey PDEs; it is a statement about the rescue mechanism in the tested reduced formulation, which is not an algebraically equivalent reduction of the full Roy PDE.

Nonlinear trade-off shapes matter because biological costs of defense need not be linear. The controlled shape grid shows that compensation can persist under selected concave, convex, and mixed endpoint-preserving trade-offs, but it also shows that branch feasibility and stability are shape-dependent. Several assumptions remain stylized: the defense trait is represented by a single frequency q, trade-offs are low-dimensional functions, q-diffusion is phenomenological, diffusion is isotropic and constant, and environmental stress enters as an added predator mortality term. These choices make the mechanism interpretable but not biologically calibrated. Empirical prediction would require species-specific estimates of defense costs, predator conversion rates, movement scales, mortality stress, and the timescale of prey evolutionary response.

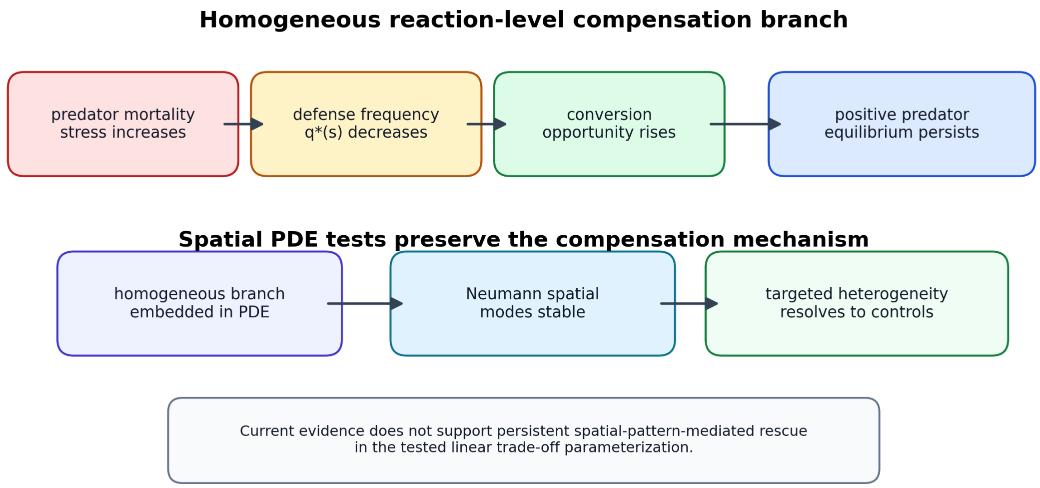

Figure 6: Final mechanism synthesis. The diagram summarizes the permitted mechanism claim: predator mortality stress shifts homogeneous eco-evolutionary balance, lower prey defense increases predator conversion opportunity, and the spatial PDE preserves the homogeneous compensation branch in the tested setup. The figure does not imply spatial-pattern-mediated rescue or biological prediction without calibration.

9 Scope and limitations

The established result in this parameterization is that the well-mixed eco-evolutionary ODE has an interior compensation branch, that the branch is feasible over an explicit stress interval, that the branch is locally stable at the tested stresses by Routh–Hurwitz criteria, and that a frozen-q counterfactual verifies indirect evolutionary rescue over a finite stress window. For the spatial PDE, the supported result is that the same homogeneous branch is linearly stable over the tested Neumann modes, sampled continuous-λ modal range, and dense sampled modal Routh–Hurwitz margin scan, and that targeted finite-amplitude heterogeneous perturbations did not provide evidence for persistent spatial-pattern-mediated rescue in the tested cases.

The controlled extension supports the presence of compensation branches beyond the linear trade-off form for selected concave, convex, and mixed endpoint-preserving shapes. This does not establish global robustness over all nonlinear trade-offs. The PDE tests are targeted rather than exhaustive. There is no global basin proof, no proof of global stability, no theorem for all heterogeneous initial conditions, and no full bifurcation analysis.

The branch equilibrium itself does not depend on the evolutionary-rate parameter ν, because the interior equilibrium condition sets the selection gradient to zero. Finite-time rescue from a specified initial condition can still depend on ν: if prey defense evolves too slowly, predators may decline before the defense shift restores predator growth balance. A supplemental finite-time diagnostic varied $\nu \in \{0.005, 0.01, 0.025, 0.05, 0.1\}$ from the pre-stress branch initial condition. At the representative rescue stress $s = 0.1191$, $\nu = 0.005$ lost the predator, $\nu = 0.01$ recovered mathematically after crossing below the extinction threshold, and $\nu \geq 0.025$ remained predator-persistent without threshold crossing under the tested horizon. This diagnostic does not alter the baseline $\nu = 0.05$ rescue-window claim.

The continuous-λ scan reduces dependence on the chosen domain for infinitesimal modes, because it samples a modal interval rather than only the discrete modes of a single rectangle. The finite-amplitude PDE perturbation results remain conditional on the 20×20 Neumann domain. Larger domains, different boundary conditions, or different dispersal closures could allow longer-lived spatial refugia or different heterogeneous basin behavior; domain-size robustness is not claimed.

The model is stylized and not calibrated to empirical predator–prey data. Parameters are used to test mechanism, not to estimate a particular biological system. The manuscript does not claim that spatial structure generally amplifies or suppresses rescue, and it does not claim that all heterogeneous perturbations decay. The results should therefore be interpreted as a mechanistic case study: they support the homogeneous compensation branch in the tested reduced free-space model, not a universal claim about spatial evolutionary rescue.

10 Conclusions

1. The tested ODE supports indirect evolutionary rescue through prey defense evolution over the deterministic frozen-q counterfactual window $0.0726 \leq s \leq 0.1656$ on the stress grid, with a threshold-safe ecological window $0.0726 \leq s \leq 0.1511$.

2. The mechanism is a homogeneous compensation branch.

3. Branch existence follows from explicit feasibility conditions.

4. The current branch is locally stable by Routh–Hurwitz criteria.

5. The spatial PDE branch is linearly stable over the tested Neumann spatial-mode range and sampled continuous-λ modal range.

6. Targeted non-homogeneous perturbations did not provide evidence for persistent spatial-pattern-mediated rescue in the tested cases.

7. Controlled nonlinear trade-off tests show that the mechanism extends beyond the linear case but remains parameter-sensitive.

Taken together, these results support homogeneous compensation as the rescue mechanism in the reduced deterministic model: predator mortality stress is offset by evolving prey composition that improves predator conversion opportunity. This claim is conditional on the tested ODE branch, sampled PDE diagnostics, diffusion and domain choices, and finite-amplitude perturbation set. The spatial PDE diagnostics did not provide evidence that persistent spatial patterning is the rescue mechanism. Empirical prediction would require calibration to a biological system plus stochastic and finite-population treatment of low-density predator dynamics.

Data and code availability

All scripts, CSV outputs, figures, and LaTeX sources are maintained in the repository at https://github.com/Thithilia/Mathbio. The external-review snapshot for this manuscript is identified by tag `external-review-2026-06-06`; the exact snapshot commit is the tag target recorded in the repository metadata. The Python environment was frozen with `python -m pip freeze` in `environment/requirements-2026-06-06.txt`, and the PDFs were built with the local LaTeX installation. No archival release DOI exists yet; the repository should be archived before journal submission.

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